MIL-C-11532C(Cm1C)
29 December 1961
SUPERSEDING
MIL-C-11532B(Cm1C)
28 March 1960

MILITARY SPECIFICATION

CASEIN

1. SCOPE

1.1 Scope. This specification covers two types of casein for use in biological media.

1.2 Classification. Casein shall be of the following types as specified (see 6.1):

Type I - Acid precipitated casein produced by a batch process.
Type II - Acid precipitated casein produced by a continuous process.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes.
PPP-D-723 - Drums, Fiber.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Products.

3.1.1 Type I. Type I casein shall be an acid precipitated product manufactured by a batch process.

3.1.2 Type II. Type II casein shall be an acid precipitated product manufactured by continuous precipitation followed immediately by continuous washing, drying and grinding.

3.2 Nutritional value. The product covered by this specification shall supply sufficient nutritional value to support adequate growth of various micro-organisms (see 6.2) when tested as specified in 4.6.1.

3.3 Pre-award sample. At no cost to the Government, each bidder shall forward a sample of the first lot of casein intended for shipment under the contract or order to the destination indicated by the contracting officer. The Government will test the samples to determine compliance with the requirements of this specification.

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3.4 Appearance. The casein shall be a granular product containing no dirt or other dark particles.

3.5 Visible contaminants. The surface of the casein in the shipping containers opened for inspection shall be completely free from insects or other arthropods, larvae and webs.

3.6 Arthropod contamination. The casein shall show no evidences of arthropods, or arthropod fragments or eggs when tested as specified in 4.6.2.

3.7 Odor. The casein shall have a characteristic but no putrescent odor when tested as specified in 4.6.3.

3.8 Chemical and physical properties. The chemical and physical properties of the casein shall conform to the requirements specified in table I when tested in accordance with the test paragraphs referenced therein.

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Table I - Chemical and Physical Properties

Property	Requirements		Test Paragraph
	Type I	Type II	
Particle size retained on 590 micron (No. 30) sieve (max.) passing 177 micron (No. 80) sieve (max.)	5% 5%	5% 5%	4.6.4
Moisture (max.)	10%	10%	4.6.5
Protein* (min.)	88%	88%	4.6.6
Fat* (max.)	1.5%	1.5%	4.6.7
Ash* (max.)	3.5%	2.5%	4.6.8
Acidity* (as lactic acid) (max.)	1.0%	0.5%	4.6.9
Reducing substances* (as lactose) (max.)	1.5%	0.75%	4.6.10
pH (min.)	4.2	4.5	4.6.11
Bacterial count (max.)	10,000 per g	5,000 per g	4.6.12
Solubility	completely soluble	completely soluble	4.6.13
Heavy metals* (as copper) (max.)	.003%	.003%	4.6.14

*On a dry basis.

3.9 Acceptance lots. The lot or lots submitted to the Government for final acceptance shall be the lot or lots from which the acceptance samples (see 4.4.2) were selected and approved by the Government as complying with the requirements of this specification.

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3.10 Storage conditions. During the period between the taking of the acceptance samples and the time of shipment, the lot or lots shall not be subjected to any known condition which may have adverse effects on the nutritional value of the casein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government responsibility. The Government will perform the tests specified in 4.6.1.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedures which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.4, 3.5, 3.9, 3.10, and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the casein offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.4 Sampling.

4.4.1 Pre-award sample. A 10-pound (lb) representative sample of the casein shall be placed in a clean, dry container labeled in accordance with Standard MIL-STD-129.

4.4.2 Acceptance sample.

4.4.2.1 For nondestructive examination. Sampling shall be conducted in

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accordance with Standard MIL-STD-105.

4.4.2.2 For test. Sample containers shall be taken at random from each lot as follows:

Lot size	Sample size
1	1
2- 275	2
276- 545	3
546- 900	4
901-1345	5
1346-1875	6
1876-2500	7

Sample sizes for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = Sample size.
N = Lot size.

A 10-lb specimen shall be removed from each container in the sample and placed in a clean, dry container labeled to identify the lot and container from which it was taken. A sufficient and uniform quantity taken from each of the sample containers shall be blended into a 10-lb composite specimen and placed in a clean, dry container labeled to identify the composite specimen with the lot represented. The composite specimen and one individual specimen shall be tested as specified in 4.6.

4.5 Examination. Sample containers of the casein shall be examined for compliance with level A and B preparation for delivery requirements in accordance with the classification of defects and with Standard MIL-STD-105.

4.5.1 Classification of defects.

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4.5.1.1 Preparation for delivery. (Section 5)

Categories and defects

Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Damaged or leaking containers	
102	Labelling incorrect	
103	Containers underfilled	
104	Contamination	
Minor:	AQL 4.0 percent defective	
201	Labelling missing or illegible	
202	Non-uniformity of container size or type	

4.6 Tests. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Where applicable, blank determinations shall be conducted in parallel with the tests using the same quantities of the reagents and corrections applied when significant. Tests shall be conducted as follows:

4.6.1 Nutritional value. The Government will combine the casein with known standard materials to form a medium and determine its nutritional value by means of the growth rate of various micro-organisms.

4.6.2 Arthropod contamination.

4.6.2.1 Solvent. Dissolve 10 grams (g) of pepsin in 500 milliliters (ml) of water, add 25 ml of concentrated hydrochloric acid and dilute to 1,000 ml with water. Mix the solution thoroughly and filter. Reject the solvent if the pH is not approximately 1.1.

4.6.2.2 Procedure. Add approximately 100 g of the specimen to 500 ml of the solvent in a 1,000 ml beaker. Stir the mixture with a high-speed mixer, transfer it to a 1,000 ml Erlenmeyer flask and heat to 40° to 45°C. Place the flask in a constant temperature bath at 40° to 45°C for 40 minutes, stirring the mixture every 5 minutes. At the end of this period, remove the flask from the bath, and boil the contents for approximately 10 minutes. Filter the solution, with suction, through a Buchner funnel fitted with a ruled filter paper designed for the recovery of extraneous matter in foods and drugs, maintaining the solution at a temperature of

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80° to 90°C, pouring small quantities of the solution on the paper at a time and washing the filter with a strong stream of water at approximately 90°C to prevent clogging of the filter paper. Examine the filter paper with a wide-field stereoscopic binocular microscope (magnification 30X) for the presence of insects, insect fragments and insect eggs.

4.6.3 Odor. Soak 10.0 ± 0.1 g of the specimen in 10 ml of water at room temperature; Add 10 ml of a 25-percent suspension of calcium hydroxide in water with stirring. Note the odor of the mixture after it has been allowed to stand for approximately 10 minutes.

4.6.4 Particle size. Assemble a 590 micron (No. 30) and a 177 micron (No. 80) sieve, both conforming to Specification RR-S-366, with the 590 micron sieve on top and fit them with a cover and receiving pan. Place 100.0 ± 0.1 g of the specimen on the 590 micron sieve and replace the cover. Shake the nest of sieves for 2 minutes by means of a single eccentric type mechanical shaker at a uniform speed of 300 ± 15 gyrations and 150 ± 10 taps per minute. Weigh the residue on the 590 micron sieve and the material in the pan. Calculate the results using the formulas:

$$\text{Percent retained on 590 micron (No. 30) sieve} = \frac{100A}{W}$$

$$\text{Percent passing 177 micron (No. 80) sieve} = \frac{100B}{W}$$

where: W = weight of specimen taken for sieve test.
A = weight of residue on 590 micron (No. 30) sieve.
B = weight of materials in pan.

4.6.5 Moisture. Weigh 3 g of the specimen to the nearest milligram (mg) in a tared glass-stoppered weighing bottle and place the bottle in an oven at 100° to 105°C for 5 hours with the stopper removed. At the end of this period, remove the bottle from the oven, place the bottle in a desiccator to cool and reweigh it to the nearest mg with the stopper replaced. Calculate the percentage of moisture, using the formula:

$$\text{Percent moisture} = \frac{\text{loss of weight (g)} \times 100}{\text{weight of specimen (g)}}$$

4.6.6 Protein. Weigh 0.5 g of the specimen to the nearest 0.1 mg and place it in a digestion flask. Add 0.7 g of mercuric oxide or 0.65 g of metallic mercury, 10 g of potassium sulfate or anhydrous sodium sulfate, and 25 ml of concentrated sulfuric acid. Place the flask in an inclined position and heat it gently until frothing ceases. (If necessary, a small amount of paraffin may be added to reduce frothing.) Briskly boil the mixture until the solution clears, and then for a least 30 minutes

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longer. Cool the solution, cautiously add about 200 ml of water, and recool the solution to below 25°C. Add and thoroughly mix 25 ml of sulfide solution consisting of 40 g of sodium sulfide or potassium sulfide dissolved in 1 liter of water, or of thiosulfate solution consisting of 80 g of sodium thiosulfate (pentahydrate) dissolved in 1 liter of water to precipitate the mercury. Add a few zinc granules to prevent bumping, tilt the flask, and add 25 g of sodium hydroxide pellets without agitation of the solution. (A sufficient quantity of a sodium hydroxide solution, consisting of 450 g of sodium hydroxide in 1 liter of water, to make the contents of the flask strongly alkaline may be used.) Immediately connect the flask to a distilling bulb attached to a condenser which has its tip immersed in 50.0 ml of 0.1N hydrochloric or 0.1N sulfuric acid contained in a 500 ml Erlenmeyer flask. Add a few drops of methyl red indicator to the Erlenmeyer flask. Rotate the digestion flask to mix the contents, and heat it until all of the ammonia has been distilled (150 to 200 ml of distillate). Titrate the excess of standard acid in the Erlenmeyer flask with 0.1N sodium hydroxide using methyl red as the indicator. Run the blank determination on 2.00 g of sugar, employing the same quantities of chemicals used for the specimen. Calculate the percent protein using the formulas:

$$\text{Percent nitrogen ("as is" basis)} = \frac{1.401\ (A-B)\ N}{W}$$

$$\text{Percent protein ("as is" basis)} = \text{percent nitrogen} \times 6.38.$$

$$\text{Percent protein (dry basis)} = \frac{\text{percent protein ("as is" basis} \times 100)}{100-M}$$

where: A = ml of 0.1N sodium hydroxide used in blank.
B = ml of 0.1N sodium hydroxide used in specimen.
N = normality of 0.1N sodium hydroxide.
W = weight of specimen in g.
M = percent moisture (see 4.6.5).

4.6.7 Fat. Weigh 1 g of the specimen to an accuracy of 1 mg and transfer it to a mojonnier fat extraction flask or a rohrig tube. Add 8.5 ml of water, and shake the mixture until it is homogeneous, warming if necessary. Add 1.5 ml of concentrated ammonium hydroxide, and heat the mixture in a water bath at 60° to 70°C for 15 minutes, shaking occasionally. Cool the solution, and add and mix 10 ml of 95-percent ethyl alcohol and 25 ml of ethyl ether. Stopper the flask or tube with a cork or with a stopper made of rubber unaffected by the usual fat solvents, and vigorously shake the mixture for 1 minute. Add 25 ml of petroleum ether (redistilled slowly at a temperature below 65°C), and vigorously shake the mixture for 1 minute. Centrifuge the mojonnier flask for 20 minutes, or allow it (or the rohrig tube) to stand until the upper liquid is practically clear. Decant the ether solution into a suitable weighing flask or metal dish. Wash the lip and stopper of the

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extraction flask or tube with a mixture of equal parts of the two fat solvents, adding the washings to the weighing flask or dish. Repeat the extraction (using 4 ml of ethyl alcohol, 25 ml of ethyl ether, and 25 ml of petroleum ether) as described above, adding the solvents and the washings to the first extract. Evaporate the solvents on a steam bath at a temperature which effects complete evaporation, but not so high that spattering or vigorous bumping will result. Dry the fat to constant weight in an oven at 100° ± 1°C. Cool the flask or dish and weigh it to the nearest mg, using a duplicate container handled in the same manner as a counterpoise, avoiding wiping either container before weighing. Remove the fat completely from the container with warm petroleum ether, dry the container and reweigh it as before. (The loss in weight is the weight of the fat.) Correct the weight of the fat by the blank determination on the reagents used. Calculate the percent fat using the formula:

$$\text{Percent fat} = \frac{\text{weight of fat (g)} \times 10{,}000}{(100 \text{ minus the percent moisture}) \times \text{weight of specimen (g)}}$$

4.6.8 Ash. Weigh 5 g of the specimen to the nearest mg in a previously tared porcelain dish. Ignite the specimen cautiously to avoid spattering, removing the burner while the casein is burning. When the casein stops burning, complete the ignition in a muffle furnace at a temperature not exceeding 550°C until constant weight is reached. Cool the dish in a desiccator and weigh it to the nearest 0.1 mg. (Save the ash for the test specified in 4.6.14.) Calculate the percent ash, using the formula:

$$\text{Percent ash ("as is" basis)} = \frac{100A}{W}$$

where: A = weight of residue (g).
W = weight of specimen (g).

4.6.9 Acidity (as lactic acid). Weigh 10 g of the specimen to the nearest mg, and place it in a 250 ml flask. Add 100.0 ml of carbon dioxide-free water, vigorously shake the mixture, allow it to stand for 1/2 hour protected from atmospheric carbon dioxide, and then filter. Titrate 25.0 ml portions of the filtrate with 0.1N sodium hydroxide using phenolpthalein as an indicator. Calculate the percent acidity as lactic acid as an average of the individual results using the formulas:

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Percent lactic acid ("as is" basis) = $\frac{9.0\ VN}{W}$

where: V = ml of 0.1N sodium hydroxide (per 25 ml portion).
N = normality of sodium hydroxide solution.
W = one-fourth of the weight of the specimen.

Percent acidity (as lactic acid) (dry basis) =

$$\frac{\text{percent lactic acid ("as is" basis) x 100}}{\text{100 minus the percent moisture}}$$

4.6.10 Reducing substances (as lactose).

4.6.10.1 Chemicals.

(a) Asbestos. Digest amphibole asbestos in hydrochloric acid (1 to 3) for 2 to 3 days, wash it free from acid, digest it for 2 to 3 days in a 10-percent aqueous sodium hydroxide solution, and treat it for a few hours with hot alkaline tartrate solution (see c). (Old alkaline tartrate solution may be used for this purpose.) Wash the treated asbestos free from alkali, digest it for several hours with nitric acid (1 to 3) and wash it free of acid. Shake the washed asbestos with water until it forms a fine pulp.

(b) Copper sulfate solution. Dissolve 34.639 g of copper sulfate (pentahydrate) in water, dilute it to 500.0 ml, and filter the solution through a mat of treated asbestos (see a). Determine the copper content of the solution (preferably by electrolysis) and adjust the concentration so that the solution contains 440.9 mg of copper per 25 ml.

(c) Alkaline tartrate solution. Dissolve 175 g of Rochelle salt and 50 g of sodium hydroxide in water, dilute the solution to 500 ml, allow it to stand for 2 days, and filter it through a mat of treated asbestos (see a).

4.6.10.2 Procedure. Weigh 10 g of the specimen to the nearest mg and place it in a 500 ml wide-mouth bottle. Add 250 ml of 50-percent ethyl alcohol, and shake the mixture for 4 hours in a shaking machine. Pipette 100.0 ml of the solution into a 200 ml volumetric flask and neutralize it with dilute acetic acid. Add 5 ml of 37-percent hydrochloric acid and allow the solution to stand overnight for inversion of the sugars. Pour the inverted sugar solution into a 400 ml beaker and neutralize with anhydrous sodium carbonate. Return the neutralized solution to the volumetric flask, dilute with water to the mark, and filter if necessary.

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Transfer 25 ml each of the copper sulfate and alkaline tartrate solutions to a 400 ml alkaline-resistant glass beaker and add a 100.0 ml aliquot of the inverted sugar solution. Cover the beaker with a watch glass and heat it on an asbestos-centered wire gauze over a bunsen burner. Regulate the flame so that boiling begins in 4 minutes, and continue the boiling for exactly 2 minutes. This rate of heating must be strictly observed. (To regulate the burner, make preliminary tests using 25 ml of each reagent and 100 ml of water before proceeding with the actual determination. An electric heater may be used instead of the bunsen burner.) Immediately filter the contents of the beaker through a previously tared gooch crucible containing a 1/4-inch mat of treated asbestos using suction. Wash the precipitate of cuprous oxide thoroughly with water at 60°C, then with 10 ml of ethyl alcohol, and finally with 10 ml of ethyl ether. Dry the crucible and contents for 30 minutes in an oven at 100°C, cool it in a desiccator and weigh. Perform the blank determination using 25 ml of each reagent and 100 ml of water, and if the weight of the cuprous oxide exceeds 0.5 mg correct the results of the reducing sugar determination accordingly.

4.6.10.3 Calculations. Use the standard Munsen and Walker tables (see 6.4) to determine the mg of lactose corresponding to the weight of cuprous oxide. Compute the percent reducing substances as lactose using the formulas:

$$\text{Percent reducing substances ("as is" basis)} = \frac{\text{g lactose} \times 100}{\text{g of sample in aliquot}}$$

$$\text{Percent reducing substances (as lactose) (dry basis)} = \frac{\text{percent reducing substances ("as is" basis)} \times 100}{\text{100 minus the percent moisture (see 4.6.5)}}$$

4.6.11 pH. Place 10.00 ± 0.01 g of the specimen in a graduated cylinder, and add carbon dioxide-free water to obtain a total volume of 100 ml. Shake the mixture until a uniform suspension is obtained. Determine the pH of the suspension at 25°C by means of a glass electrode-saturated calomel-pH meter system (see 6.5) in accordance with the instructions on the meter.

4.6.12 Bacterial count. Use aseptic techniques throughout this test. Weigh 1 g of the specimen to an accuracy of 1 mg, and triturate it in 5 ml of water. Add the triturate to 95 ml of water and shake the mixture 25 times through a 1 foot arc. (This is a 1:100 dilution.) Place a 1.0 ml portion of the 1:100 dilution on each of three nutrient agar plates, using a standard plate count technique. Incubate the plates at 30°C for 48 hours, and then count the number of colonies on each plate, using a colony counter.

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Calculate the bacterial count per g of the dry casein by multiplying the average colony counts of the three plates by 100.

4.6.13 Solubility. Place 5.00 ± 0.005 g of the specimen in a 250 ml Erlenmeyer flask containing 75 ml of water at a temperature of 47° ± 2°C. Thoroughly shake the flask and allow it to stand for 10 minutes. Add 0.80 g of sodium tetraborate (borax), and shake the mixture for 30 minutes at a temperature of 47° ± 2°C. Remove the flask from the shaker and visually examine the contents for evidences of undissolved material.

4.6.14 Heavy metals (as copper).

4.6.14.1 Chemicals.

(a) Hydrogen sulfide solution. Use a freshly prepared saturated solution of hydrogen sulfide in water.

(b) Copper standard solution (0.01 mg copper per ml). Dissolve 0.0393 g of cupric sulfate (pentahydrate) in sufficient water to make 1,000 ml of solution.

4.6.14.2 Procedure. Add 3 ml of concentrated hydrochloric acid and 1 ml of concentrated nitric acid to the crucible containing the residue from 4.6.8, and evaporate the mixture to dryness on a steam bath. Take up the residue with 30 ml of water and 0.5 ml of 1N hydrochloric acid and pour the solution into a match comparison tube. (Filter the solution, if necessary, using wash water sparingly.) Add 10 ml of hydrogen sulfide solution, dilute the mixture to 50 ml and shake. Prepare a standard by mixing 15 ml of copper standard solution, 0.5 ml of 0.1N hydrochloric acid, 10 ml of hydrogen sulfide solution, and sufficient water to make 50 ml of the mixture in a second matched comparison tube. Make observations of the comparative colors of the contents of the comparison tubes, viewing them lengthwise through the tubes against a white background. (If the sample solution is darker than the standard solution, the casein contains more than 0.003 percent copper.)

4.6.15 Acceptance/rejection criteria. If either the composite specimen or the individual specimen of casein fails to meet the requirements of this specification, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Casein shall be packed in a fiber drum with metal heads, conforming to the requirements of Specification PPP-D-723, type III, grade A. The closure shall be sealed with a rubber gasket or sealing disk.

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Materials used shall be compatible with and shall in no way affect the casein.

5.1.2 Level B. Casein shall be packed in a fiber drum conforming to the requirements of Specification PPP-D-723, type II, grade A. Closure shall be as specified in 5.1.1.

5.1.3 Level C. Casein shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of casein required.
(c) Level of packing required.
(d) That pre-award samples will be required.

6.2 Nutritional value. Due to unknown factors affecting the growth of various micro-organisms and the possible danger in handling these organisms, the Government will conduct all tests on samples to establish the nutritional value of the product. It is estimated that within 30 days after receipt of the pre-award samples or acceptance samples the prospective bidders or supplier will be notified of the acceptance or rejection of the product on the basis of nutritional value.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing process intended to make the final product substantially uniform.

6.4 Munsen and Walker Tables. The Munsen and Walker Tables for the determination of lactose from reduced copper may be found in:

(a) Official Methods of Analysis of the Association of Official Agricultural Chemists.

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(b) Handbook of Chemistry and Physics, Chemical Rubber Publication 31st Edition, page 1470.

(c) Journal American Chemical Society, Volume 28, page 663 (1906).

6.5 pH meters and systems. pH meters and systems complying with the requirements of ASTM Designation E70 have been found to be satisfactory for pH determinations.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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